Proposal for ParkFinder Website

Problem Statement

Despite the growing importance of green spaces in urban environments, many individuals face challenges when trying to discover suitable parks that meet their specific needs. Existing platforms such as Google Maps or TripAdvisor offer general information about locations but lack dedicated, community-driven tools tailored for park discovery and engagement. Users often struggle to find comprehensive and up-to-date information about park amenities, accessibility, and visitor experiences. Furthermore, there is no streamlined platform that allows users to contribute their insights, share experiences through blogs, and participate in an active community centered around park exploration.

Additionally, administrators of such platforms often lack efficient tools for managing user-generated content, responding to inquiries, and distributing updates through newsletters. This leads to outdated or irrelevant content, decreased user engagement, and missed opportunities for promoting underutilized public parks.

To address these challenges, a centralized, scalable, and user-friendly web platform is needed — one that not only enables users to find parks based on location but also supports user interaction, experience sharing, and administrative content moderation. ParkFinder aims to fill this gap by delivering a specialized, Golang-powered solution optimized for performance, community contribution, and effective administration.

Abstract

ParkFinder is an innovative web-based platform designed to help users discover parks in their location while allowing them to share their experiences through reviews and blogs. The system facilitates user authentication, park listing, and an interactive reviewing process. Administrators have additional functionalities for content moderation, contact management, and newsletter distribution. Built using Golang, ParkFinder provides an efficient, scalable, and secure solution with server-side rendering for optimal performance. This proposal outlines the project’s objectives, literature review, features, technical specifications, and development roadmap.

Introduction

Parks play a significant role in urban planning, offering recreational spaces for communities. However, finding a suitable park with the desired amenities can be challenging. ParkFinder aims to bridge this gap by providing a digital solution where users can locate parks, read reviews, and contribute their experiences.

Literature Review

Importance of Parks in Urban Spaces

Parks serve as vital community spaces for relaxation, fitness, and social interactions. According to research, urban parks contribute to improved mental health, reduced air pollution, and increased community engagement. However, limited information accessibility about park locations, facilities, and reviews often discourages people from visiting them.

Digital Solutions in Outdoor Recreation

Several platforms provide location-based services, such as Google Maps and TripAdvisor, which offer reviews for general locations. However, a dedicated platform focusing specifically on parks and user-generated content is lacking. ParkFinder provides a specialized service where users can not only locate parks but also share detailed experiences and recommendations.

Use of Golang in Web Applications

Golang is known for its high performance, concurrency, and efficient memory usage, making it suitable for web applications requiring fast data processing. By implementing Golang, ParkFinder ensures seamless database transactions, template rendering, and API integration while maintaining a lightweight server-side architecture.

Objectives

1. To develop a digital platform for users to find parks based on their location.
2. To provide a platform for users to create accounts and share their park experiences through blogs and reviews.
3. To implement an administrative panel for managing user content, newsletters, and user interactions.
4. To ensure seamless user authentication, content moderation, and secure data management.
5. To optimize the platform for scalability, security, and high-performance rendering.

Features & Functionality

User Features

- **User Authentication**: Users can create accounts, log in, and manage profiles securely.
- **Park Listings**: Users can search for parks by location, create new park entries, and update existing ones.
- **Park Reviews**: Users can write and read reviews, providing insights into various parks.
- **Newsletter Subscription**: Users can subscribe to newsletters for park updates.
- **Contact Form**: Users can send inquiries via a "Contact Us" form.

Administrator Features

- **User Management**: Admins can create, view, authenticate, and manage users.
- **Park Management**: Admins can create, update, and delete parks.
- **Review Moderation**: Admins can delete inappropriate reviews.
- **Newsletter Management**: Admins can create and handle newsletters.

- **Contact Requests Management**: Admins can manage and respond to inquiries.

Technical Specifications

Backend

- **Language**: Golang (Go)
- **Framework**: Custom-built template engine for UI rendering
- **Database**: MySQL with structured domain-driven data models
- **API Integration**: AJAX calls for dynamic content updates

Frontend

- **Templating Engine**: Server-side rendering using Go’s built-in templating
- **AJAX Integration**: Used for reviews, contact forms, and newsletter management

Deployment

- **Port Configuration**: Runs on port 4000 for HTTP and 40001 for HTTPS
- **Execution Modes**:
  - Golang runtime: `go run server.go`
  - Windows executable: `parkfinder.exe`

Development Roadmap

1. **Database Setup**
   - Develop domain and database functions.
   - Implement structured schema for users, parks, reviews, and admin functionalities.
2. **User Management**
   - Implement authentication and authorization features.
   - Enable user registration, login, and profile management.
3. **Park Management**
   - Develop park listing functionalities based on location.
   - Allow users and admins to create and update parks.
4. **Review System**
   - Implement AJAX-based review submission and management.
   - Provide review filtering and rating systems.
5. **Administrative Panel**
   - Develop content moderation tools for parks and reviews.
   - Implement newsletter and contact request management.
6. **Testing & Deployment**
   - Perform system, security, and usability testing.
   - Deploy the platform to a production environment.

Conclusion

ParkFinder is a scalable, efficient, and user-friendly platform that enhances park discovery and user engagement. By leveraging Golang’s speed and performance capabilities, ParkFinder ensures a seamless experience for users and administrators. This proposal outlines the development roadmap, technical specifications, and features required to achieve the project's objectives.